

White Paper: Proof of Social Learning Protocol

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Abstract

This white paper introduces a novel Proof of Social Learning Protocol (PSLP) designed to transparently and immutably record knowledge transfer and learning events within communities of practice, using Distributed Ledger Technology (DLT), specifically a blockchain. Unlike traditional blockchain protocols that rely on computational proof-of-work for validation, PSLP leverages social learning events as the fundamental validation mechanism ("mining").

This revision frames the protocol in terms of **organizational games**: structured interactions — of which the Liberating Structures are the most mature and best-refined example — that deliberately organize a group's time, attention, information processing, and energy. Two central commitments distinguish this version. First, the protocol records not only *what* a community did, but **how the group's thinking moved** — it accumulates **traces**, not bare agreements, so that the community can see and sense its own system. Second, the protocol's value is **syntropic**: it favors structures and practices that become more organized and more ordered over time, expanding the possibility space of the communities that use them rather than dissipating their potential.

The protocol is particularly well-suited to communities focused on procedural knowledge, such as the Liberating Structures community of practice, where specific methods or "structures" can be seen as discreet packets of procedural knowledge. A core component is a shared Learning Journal (wiki) interface that uses recorded learning progress, community evaluation, and available data to enhance user learning and facilitate minimally resource-intensive information organization. Privacy, data sovereignty, security, ethical sharing, attribution, and the cultivation of a regenerative economy of value, discovery, divestment, and distribution are paramount to the protocol's design.

1. Introduction: The Value of Capturing Social Learning

Knowledge transfer and learning within communities are often implicit, fragmented, and difficult to track or formally recognize. While digital platforms exist for sharing information, they frequently rely on centralized models that can lead to silos, data exploitation, and a lack of trust. Traditional credentialing systems can be costly, exclusive, and may not accurately reflect embodied practice or collaborative learning.

1.1 A capacity that is real but unclaimed

Much of the most valuable work that happens inside groups is **organizational design in the broadest sense**: the responsible organization of a group's time, attention, information processing, and energy. Human attention is the scarcest and most valuable asset any group possesses, and the person who can reliably structure it — who can set up the conditions under which a group thinks well, decides well, and acts together — is exercising a capacity that is widely needed and rarely credibly evidenced.

We call the holder of this capacity an **organizational game designer**: a practitioner who designs *games* in the deepest sense — socially constructed rule systems for distributing attention and power — and who has the demonstrated skill of deploying them in specific contexts. Liberating Structures practitioners, skilled facilitators, experienced coaches, and capable meeting chairs are all working instances of this capacity. The Liberating Structures literature is itself testimony to its value: structures are "incredibly well-designed," and by leaning into their design elements — how participation is distributed, how groups are formed, the steps over time, the invitation, and the organization of materials and space — a practitioner can understand how group intelligence actually works and how to work with it.

Three problems attend this capacity today:

1. **It is implicit.** It lives in embodied practice, in the design decisions behind a meeting, in the *system of justification* the practitioner holds when asked why they chose these elements and not others. It is, in the words that describe the field itself, "super easy to use, hard as anything to explain."
2. **It is unevidenced.** There is no shared record of who has demonstrated it, in what contexts, at what scale, with what results. Claims are cheap; evidence is costly, and the evidence is currently nowhere to be found.
3. **It is unvalued in formal terms.** Credentialing that would make this capacity *declarable* — a non-speculative, community-grounded way of stating "I can do this" — does not exist for the general practitioner.

Capturing and valuing social learning requires new approaches that align with the principles of decentralized collaboration, community empowerment, and the recognition of diverse forms of value beyond the purely monetary. This white paper proposes the Proof of Social Learning Protocol (PSLP) as a framework for building such a system — one that treats the social learning events of a community of practice as the primary evidence of the capacities those events produce and refine.

1.2 Why a distributed ledger

Blockchain technology offers a decentralized and transparent means of recording transactions and data. However, many blockchain applications have focused primarily on financial value or speculative assets, often reproducing existing inequalities and power dynamics. There is a need to explore how DLT can be applied to support social and community-driven objectives, moving beyond purely transactional exchanges to enable social and solidarity practices. PSLP is one such application: a ledger whose native "currency" is not tokens but **attested evidence of learning and capacity**.

2. The Liberating Structures Community: A Suitable Use Case

The Liberating Structures (LS) community of practice provides an ideal context for piloting a Proof of Social Learning Protocol. LS are a set of forty-three microstructures that can be used to engage groups of any size in collaborative problem-solving, innovation, and planning. They are simple rules to unleash a culture of innovation and make it practical for people at any skill or hierarchical level to quickly become expert contributors.

Importantly, each Liberating Structure is a **discreet packet of procedural knowledge**, defined by five design elements: how participation is distributed, how groups are formed, the steps over time, the invitation (question/prompt), and how materials and space are organized. Learning LS involves direct experience, practice,

and debriefing with peers — not instruction alone. As the field guide puts it: you will not believe what the structures do for you *until you have practiced the methods and learned from them*. This has two consequences for the protocol, developed in §3.2 and §3.6:

- Every LS practice session is **embodied skill acquisition**, and its record is strong evidence of practitioner capacity.
- The community that practices the structures is simultaneously **training the validators** of any protocol built around them.

The LS framework and its community embody key principles that align with PSLP's goals:

- **Focus on Process:** LS emphasizes how people interact and collaborate, providing a structured approach to social dynamics.
- **Community of Practice:** Learning and spreading LS occurs through shared experiences and peer-to-peer interaction.
- **Open Source Nature:** LS is licensed under Creative Commons, encouraging adaptation and sharing.
- **Diverse Participation:** LS methods are designed to include and unleash everyone, regardless of hierarchy or skill level.
- **Design discipline:** LS are "incredibly well-designed" and carry explicit design elements and a system of justification — which makes them natural, structured units of record on the ledger.

Documenting the application and learning of these structured interactions provides a tangible starting point for recording complex social learning events on a ledger.

3. The Proof of Social Learning Protocol (PSLP)

3.1 Core Concept: An Evidence Archive, Not a Transaction Log

The PSLP utilizes a DLT, specifically a blockchain, as an open, shared, decentralized database to record information about social learning events. This section restates the core concept with the emphasis this revision places on it: **the ledger is best understood as an evidence archive** — a growing, permissioned, community-validated record of what people have actually done, in what contexts, with what attested outcomes.

In a commons, **claims are cheap and evidence is costly**. Anyone can say they are a capable facilitator; the ledger accepts only what participants have collectively attested. Each recorded social learning event is therefore an *attested observation about the capacities of the people involved*, not a self-declaration. No single record certifies anyone. Credential emerges from the **pattern across records**: structures used, contexts of application, scale of groups, novelty of situations, and the community evaluation recorded in the Reflection field (§3.3).

Equally important is what the records *are*. A meeting minutes document is an **agreement** — a snapshot of consensus at a point in time. A PSLP record is a **trace**: a record of motion, of how a group's thinking actually moved through an event — what it touched, what it produced, what was learned in the doing. The distinction matters because traces are what you can look back on and *sense the system* through. The protocol's native data unit is therefore the trace, and the Learning Journal (§3.4) is the place where traces accumulate until the community's own system becomes visible to itself.

This ledger provides a transparent and immutable record of learning activities within the community — a record whose value grows with every event, because every event adds not just information but **capacity evidence** about the community's own designers.

3.2 Social Learning Events as "Mining"

Instead of relying on energy-intensive computational puzzles for validation, as in Proof-of-Work, PSLP uses social learning events themselves as the "mining" process that validates and adds data to the ledger. A social learning event could be:

- A practice session where a specific LS is used.
- A workshop or training where LS methods are taught and practiced.
- A coaching session applying LS-based inquiry.
- Any structured interaction where explicit learning or knowledge transfer using defined protocols occurs.

During or after a social learning event, participants collectively, or designated stewards/facilitators, submit a record of the event to the network. The validation of this record is based on the social consensus of the participants involved, potentially combined with community-based attestation or evaluation processes. For example, participants could cryptographically sign off on the record of the event, or a quorum of participants could attest to its occurrence and basic details. This distributed trust mechanism validates the "block" of learning data before it is added to the chain. This aligns with the idea of distributing trust rather than removing it entirely.

First-class events. While all structured learning interactions qualify as social learning events, a subset is *first-class*: events in which participants practice the community's own method. Training and practice in the Liberating Structures themselves is the paradigmatic case, because LS are acquired through direct experience, practice, and peer debriefing rather than instruction alone. Such records are first-class in two ways:

1. **They are the strongest evidence of practitioner capacity.** A participant who has structured, run, and debriefed a practice session has demonstrated the very capacity the community values above others — the responsible organization of a group's time, attention, and energy processing.
2. **They train the protocol's own validators.** The quality of the ledger depends on the capacity of those who witness, attest, and record its blocks. Practice-in-the-method records document exactly that capacity in formation. The ledger thus records the events that produce its qualified validators — a productive self-reference, since the evidence is attested by participants rather than issued by an external authority.

First-class records carry the same data model as any other event (§3.3) but are flagged as such, and they form the primary evidence base for the capacity certification described in §3.6.

3.3 Data Model: Recording Learning Activities

Each social learning event recorded on the PSLP ledger would include structured data, potentially captured through intuitive interfaces to minimize user effort and resource intensity. The data model would encompass:

- **Context:** Where and when the learning event took place (e.g., organization, community group, date, time).
- **Objective:** The stated purpose or goal of the learning event. For LS applications, this could link to the "Nine Whys" of the work, or to the ideal-state goal-setting practiced in facilitation traditions (the "definition of awesome" of Toyota Kata).

- **Participants and roles:** Pseudonymized or verified identities of individuals who participated, **with role distinctions:** the *designer* (the person who organized the group's time, attention, information processing, and energy processing capacity), the *stewards* (who captured and submitted the record), and the *players* (participants in the game itself). Permission layers can manage the visibility of this data based on user preferences and community norms. Distinguishing roles is what makes the ledger a capacity archive rather than an attendance log.
- **Process/Structure:** Details about the specific LS or sequence of structures ("strings") used, including time allocation and group configurations — the five design elements made explicit.
- **System of justification:** *Why these elements, and why this design?* The design rationale the practitioner holds — how the chosen structure serves the stated objective, given the participants and constraints. This field is what converts a record from "what happened" into "how a designer thinks," and it is the single most discriminating field in the model.
- **Content/Outcomes:** Summaries of discussions, key insights, emergent ideas, decisions made, or planned next steps. This could include links to external documentation or artifacts where appropriate.
- **Traces and artifacts:** The concrete artifacts the event produced — maps (e.g., Ecocycles), traces (e.g., simple ethnography notes, 25/10 crowd-sourcing outputs, "what I need from you" lists). These are the evidence that the group's thinking actually moved, and they are what later communities can read to *sense the system*.
- **Reflection:** Evaluation of Return on Time Invested, structure choice, design, novelty/innovation, second thoughts, Tips and Traps — and, per §3.7, an assessment of the event's **syntropic quality:** did the event leave the group more organized, more capable, and with a larger possibility space than it found them?

Capturing this structured data allows for the mapping of how different structures are applied in various contexts for specific purposes, and the outcomes they generate.

3.4 The Shared Learning Journal (Wiki Interface)

The shared Learning Journal serves as the primary interface for users to access and interact with the data on the PSLP ledger. Designed as a collaborative wiki, it visualizes the recorded learning events and offers functionalities to support user learning and community knowledge building. The journal leverages the structured data on the ledger in several ways:

- **Learning Progress:** Users can track their own participation in learning events and visualize their experience with different LS patterns, potentially using frameworks like the Ecocycle or developmental pathways. Because the journal is a trace archive, a user's history is not a count of events but a legible **progression of demonstrated capacity:** structures used, contexts of application, scale of groups, community-evaluated results. This progress can influence access permissions within the journal and Reflections of Social Learning Events.
- **Novelty and Utility:** The journal can highlight instances where structures were used in novel contexts or situations, or where community evaluation indicates high utility or impact.
- **Organization and Navigation:** The wiki structure allows for organizing information about LS patterns, strings, tips, traps, riffs, variations, and examples based on the recorded data. It can use metadata (context, objective, participants) to facilitate searching and filtering, enabling users to find relevant examples or connect with practitioners.

- **Seeing and sensing the system:** Because the journal accumulates *traces* rather than bare agreements, the community can watch its own system develop — which structures travel, which designers are repeatedly attested, which contexts produce which outcomes. The journal is, in effect, a sociological instrument pointed at the community's own learning.
- **Minimally Resource Intensive:** By drawing structured data from the ledger, the wiki can automate certain organizational tasks, reducing the manual effort required for content management compared to traditional wikis. Non-intrusive interfaces for data input during learning events could also minimize participant burden.

The journal fosters a knowledge commons where collective wisdom is shared and made accessible. It can be enhanced by community evaluation features, allowing users to rate or comment on the documented events or the insights derived from them, further supporting the organization and highlighting of valuable information.

3.5 Permissions and Access Control

PSLP incorporates a granular permission system, managed on the ledger and reflected in the Learning Journal, to ensure data sovereignty and control over who can access and share information. Permissions are dynamically managed based on factors such as:

- **User Identity:** Individuals control their own data, deciding which information is publicly shared, shared within specific community circles, or kept private.
- **Learning Progress/Role:** As users gain experience or take on specific roles within the community (e.g., facilitator, steward), they may be granted permissions to access or contribute to different levels of information in the journal, such as design details or aggregated data. In this revision, role is not a tag but a **data-grounded progression** (e.g., novice → practitioner → steward → certified organizational game designer), each step supported by ledger records — practice events completed, structures used, contexts of application, community-evaluated Return on Time Invested — and therefore visible, checkable, and contestable by the community.
- **Community Evaluation/Trust:** Reputation or trust built through participation and positive community evaluation can unlock higher levels of access or influence over content organization.

This ensures that while the core record of learning events is immutable on the ledger, the accessibility and visibility of detailed information within the user-facing journal are governed by community norms, trust, and individual consent.

3.6 Ethical Sharing, Attribution, and the Economy of Value

The PSLP is designed to promote ethical sharing and proper attribution of contributions. By recording the participants and process of learning events, the protocol provides a basis for crediting individuals and groups for their roles in knowledge creation and transfer.

The protocol aims to cultivate a **regenerative economy of value, discovery, divestment, and distribution** within the learning community. This goes beyond traditional monetary exchange, recognizing and valuing diverse contributions such as:

- **Value Creation:** Documenting the successful application of structures or the emergence of new insights acknowledges the value created through collaboration.

- **Discovery:** The journal facilitates the discovery of effective practices, novel applications, and potential collaborators.
- **Distribution:** Insights, knowledge, and potentially related resources or opportunities can be distributed throughout the network.
- **Divestment:** While not focused on financial divestment, the system could, for example, be used to distribute financial resources throughout the community of practice.

The four flows above describe *what* circulates in the community. The subsections that follow specify *how* value is credibly claimed, evidenced, and transferred — the mechanism by which attribution becomes something participants can rely on, not merely something the system displays.

3.6.1 Claims are cheap; evidence is costly

In a commons, claims are cheap and evidence is costly. Each recorded social learning event is an attested observation about the capacities of the people involved, not a self-declaration (§3.1). The ledger is an **evidence archive**: what people have actually done, in what contexts, with what attested outcomes.

Events recorded under §3.3 attribute distinct roles, and each role is a distinct form of evidence:

- **The designer** — the person who organized the group's time, attention, information processing, and energy processing capacity; in the protocol's terms, the organizational game designer of that event.
- **The players** — whose engagement, contributions, and reflections evidence their participation in the learning.
- **The stewards** — who captured and submitted the record, attesting to the event's occurrence and basic details.

The pattern across records — not any single record — is what makes a capacity claim credible.

3.6.2 Reputation first — and the certification that reads from it

The protocol leads with **reputation**, not certification. A practitioner's reputation is the *accreting, emergent track record* of what they have done and what others have attested — the pattern across records described in §3.6.1. It is always present, it is generated by play, and **it is never sold**. This is the honest, commons-compatible core: it requires no purchase, no licensing body, and no fee.

Certification is then the optional formalization that reads from reputation. A key design commitment: **certification does not create capacity; it formalizes it.** In conventional credentialing the certificate precedes and enables the work; the PSLP inverts this. A credential is *granted by a user group, scoped to that group, and activated by demand* — the declarable statement that the ledger now supports: *this person's demonstrated capacity meets a community-defined threshold, in this community.*

Three separations make this work — and they deliberately split what a fully-local or fully-centralized scheme would fuse:

1. **The criteria are shared — one developmental path for everybody.** Ironically for a protocol built on trust over hierarchy, the *criteria* for certification are centralized: not a local menu, but a single shared rubric that every practitioner walks the same way. The point is comparability — to have "some degree of expected competence," and to be able to "know where the blind spots and competence are." This is a standardization of the *map of competence*, not a re-centralization of authority (the seal below stays distributed). The rubric is

the deck's **Development Phases** ladder, and it is **multi-dimensional**, not a session count: - **Structures in use** — 0 (Despairing Cynic, Unconscious Incompetence) → 5 (Cautious Optimist, Conscious Naive) → 15+ (Rapturous Super-User, Conscious Competence) → 33 of 43 (Maestro Minimalist, Unconscious Competence); - **People reached** — the breadth of *distinct* practitioners and groups the work was actually done *with*, not the number of sessions delivered; - **Diversity of settings** — the range of contexts (corporate, community, family, cross-cultural, conflict, celebratory) in which the structures have been run; - **Design-goal achievement** — the degree to which the practitioner *effectively accomplished their design goals*. We are "not using these things in a vacuum": much group work is legitimately about vibes, trust, and getting to know one another, and those are real design goals — but they count only *as intended, designed toward, and effectively accomplished*. Credibility is claimed against goals met, not against time spent. A practitioner is *declared* at a phase, not assigned a tag.

2. **The seal is per user group — decentralized granting.** The *seal* is not centralized. The **user group is the certifying body**, and its **attestation is the seal** — "the thing required." The certifier grants "by the witness of your practice, or by your participation in my practice space," under their own street cred: *all right, you know this — go forth*. The community — not the platform, and not the deck author — issues the seal; **the deck author holds no standing seat in the granting**. The criteria (item 1) say *what* counts; the user group's attestation is the *proof* that it is true, for that group.
3. **Activation is demand-driven.** Certification becomes useful and hireable when there is demand — a hiring request, a client's brief. Until then it rests as a reading of reputation.

Two further commitments keep it from becoming a cash grab:

- **The pay-to-win path is structurally unavailable.** Because the evidence trail is the reputation and the credential only reads from it, there is no shortcut to purchase. What is purchasable is *assistance in developing the capacity*, not the claim of having it (§3.6.3).
- **The Creative Commons constraint is load-bearing.** Because the underlying Liberating Structures are Creative Commons, *selling access to a credential over them is not available by right*. A practitioner may point at a **proof-of-participation artifact** — evidence that justifies, for example, a corporate L&D budget — but that artifact is *proof*, not a paid license. The protocol's market value therefore *grows with evidence, not with purchase*.

3.6.3 A legitimate market layer

The protocol's commons does not require the exclusion of market exchange; it requires that exchange be structured so that it does not extract from the commons. Three market models are therefore legitimate:

- **Pay-to-learn:** A participant with more money than time may pay for design assistance — coaching, structure design, practice scheduling — that makes their learning journey more effective. The payment buys *development of capacity*, and the resulting evidence enters the ledger in the same form as evidence produced without payment.
- **Pay-to-play:** Where an event has costs (materials, space, travel), participants may contribute toward them. Distribution of such funds is itself one of the divestment flows of §3.6.
- **Pay-to-certify:** A practitioner may pay a community-set fee to have their demonstrated capacity formalized as a credential (§3.6.2). The payment funds the commons — the divestment flow — not the certifier's profit. This is the protocol's intended economic engine: the cash grab that goes to the community of practice.

All three models are constrained by an **omni-win requirement**: the transaction structure must be designed so that all parties' capacities grow — the payer's, the payee's, and the community's. A facilitation engagement that leaves the payer more capable and the payee more capable is omni-win; one that merely transfers attention from one party to the other is not. The Liberating Structures repertoire is designed to satisfy this constraint by construction: structures are simple enough that the *process of the engagement itself* builds capacity in everyone present, including the payee. Delivery of paid practice should therefore be structured **many-to-many** wherever possible — one design serving many learners — rather than one-to-one extraction.

3.6.4 First-class events: practice in the protocol's own method

A social learning event is first-class when it directly trains the capacities the protocol depends on. Practice and training in the Liberating Structures themselves is the paradigmatic case.

The field guide is explicit about how structures are learned: through direct experience, practice, and peer debriefing — not instruction alone. Every LS practice session and training is therefore embodied skill acquisition, and its record on the ledger directly evidences the development of the attention-designing capacity that §3.6.2 certification formalizes.

This makes the protocol self-referential in the productive sense: **the ledger records the events that train its own validators**. The validator is trained by the protocol's own subject matter; the credential rests on the protocol's own evidence. No external credentialing body is needed, because the ledger *is* the evidence archive. The recursion is sound precisely because the evidence is attested by participants rather than issued by an authority.

Intended audience and the value-first principle. The protocol's primary users are **user-group organizers and practitioners out in the field** — people already running these structures. Everything the protocol does — collection, attestation, certification — is in service of one first obligation: **making their life easier designing and developing their interventions**. The protocol delivers that value first, and transparently; in exchange, it *creates together* with them the repository of knowledge — a commons whose value is significantly greater than the sum of its parts, received as a co-benefit. The value delivered to the field must be *smooth, transparent, and overwhelming*; the knowledge repository is what the field builds as the byproduct. Collection is never the point of the protocol. Usefulness is.

3.6.5 Trust propagation and network topology

Attribution cannot remain local. Evidence about a facilitator's capacity in one community should inform trust in another — with appropriate attenuation. A record witnessed directly by a participant is stronger than the same record as reported by a second party, who in turn is attested by their own witnesses; trust therefore propagates through the social graph with decay across distance, and with confidence weighted by the reliability of the intermediaries carrying it.

This is fundamentally a **small-world propagation problem**. Communities of practice have the topology small-world networks describe: high local clustering (tight-knit practice groups that know their members well) and short average path lengths (a small number of bridge figures — stewards, certified designers, cross-community facilitators — connect most of the network). Evidence needs no more than a few hops to reach almost any community; the task of the protocol is to let it propagate *with* its integrity metadata intact, and to let hub nodes be recognized by the community rather than assumed by the platform.

[RESOLVED 2026-09-28 — by Mujo's Kitestring proposal, §07. The small-world gap is not merely conceptual: in the NAOMS implementation, all the trust primitives exist (signed chains, co-signing by role, FROST hive credentials, selective-disclosure R-Cards) but nothing writes trust edges — the two functions that would have no production caller, and trust propagation returns zero for every pair. So trust-weighted discovery between communities is inert until credentials are wired into trust edges. The fix — write trust edges from accepted, counter-signed credentials; feed each contact's trust flow from propagation; verify the credential behind each reference — is a one-to-two-week change that benefits every NAOMS app. Until it lands, only yes-or-no recognition between communities is honest, which is exactly the attenuated, hop-decayed propagation this subsection describes.]

Collection and presentation render independently. The application that does the registration, attestation, and evidence collection is *not* required to be the application that presents the public journal. The collection side feeds a presentation that is separately designed — more cleanly organized, usable across skill levels, and better than the existing community website at showing the state of the community of practice and how its patterns are actually used. This separation is deliberate: the ledger's job is evidence; the journal's job is knowing.

3.6.6 The regenerative character of the economy

The regenerative character of this economy is not a metaphor. Each recorded event produces more evidence than it consumes: the attestation required to record an event is itself a community practice that builds the very capacity the protocol is documenting. The currency of the network is not a token but **capacity made visible** — attributable, checkable, and, when its holder wishes, marketable. Attribution built on social consensus and community evaluation is what turns the four flows — value creation, discovery, distribution, divestment — from a description of values into a working protocol. Its thermodynamic signature is described next.

3.7 Syntropy: The Protocol's Organizing Principle

Entropy is the tendency of closed systems toward disorder and lost possibility: energy dissipates, potential flattens, and the number of things a system can do shrinks. The classic image is water flowing downhill — held high on the landscape, it can turn mills and feed organisms across its slow descent; once it reaches the sea, that entire category of possibility is gone, no longer up for grabs.

Syntropy — the term established by Lofredo de Matteo in 1945 — names the opposite capacity: the ability of open, energy-importing systems to build *local* order over time, against the default drift. In ecological practice, the image is "holding the water high in the drainage." Vegetation is literally the cellulose gauze of nature, holding water high in the landscape so that, as it moves slowly, it touches more surfaces, feeds more organisms, and — at scale — powers the biotic pump: vegetation spores clouds, clouds create rain, and the possibility space for more life *expands*. A syntropic system does not merely resist decay; it **increases the possibility space of its surroundings** by maintaining and expanding order.

The organizing goal of an organizational game is syntropic. A well-designed structure — the design brief that frames the context, the storyboard that organizes the intervention, the structure itself that carries the play — is a device for creating what we call **syntropic flows**: flows of attention and information that, as they move, leave the group *more* organized and *more* capable than they found it. Organization, in this sense, is not control imposed on a group; it is the group's own thinking being given beneficial flows — a collective metacognitive move in which the

group is thinking about an entire group's thinking, and designing the conditions under which emergence has a higher probability of occurring. Emergence is not the absence of structure; it is structure designed to make emergence more probable.

The PSLP adopts syntropy as its **measurability criterion** in three places:

1. **Per-event:** The Reflection field (§3.3) asks whether the event was syntropic — did it leave the group more organized, more capable, with a larger possibility space than it entered? Community evaluation supplies the attestation.
2. **Per-practitioner:** A practitioner's evidence pattern is syntropic when, across their recorded events, the groups they have designed for show growth in capability and optionality over time — the practitioner is *holding the water high* for the communities they serve.
3. **Per-community:** The Learning Journal, by accumulating traces, functions as a **knowledge biotic pump**: it holds the community's knowledge "high in the drainage" — visible, referenceable, reusable — so that it can flow through more practitioners, more contexts, and more generations of events than a single meeting could have touched. Each trace read by a new practitioner is water touching a new surface. The journal is the infrastructure by which a community's learning creates more learning than it consumed.

A PSLP learning event is *regenerative* in exactly this syntropic sense: it is a structure that, well-played, leaves the community more organized, more ordered, and with a larger possibility space than it entered with. That is checkable, and it is attributable — which is why the protocol's economy of value is regenerative rather than merely aspirational.

3.8 Privacy, Data Sovereignty, and Security

Ensuring privacy, data sovereignty, and security is critical. The inherent transparency of blockchain must be balanced with the need for confidentiality and user control.

- **Privacy:** While transaction details (records of learning events) are on-chain, sensitive data can be managed off-chain with references on the ledger, or encrypted using techniques like Zero-Knowledge Proofs (ZKPs) or Trusted Execution Environments (TEEs) where feasible. Pseudonymity or anonymization of participant data can also be employed. The system of justification field is naturally shareable in summary; the underlying artifacts and reflections can be permissioned in layers.
- **Data Sovereignty:** Users maintain control over their personal learning data and identity, deciding how it is shared and used in accordance with their preferences and community agreements. This aligns with the principle of "nothing about us without us."
- **Security:** The DLT provides inherent security through cryptographic proof and distributed consensus, protecting the integrity and immutability of the learning records. Robust access controls based on cryptography can protect sensitive information.

Developing this aspect requires careful design, potentially leveraging advancements in privacy-preserving technologies within the DLT space.

4. Challenges and Future Work

Implementing a Proof of Social Learning Protocol presents several challenges:

- **Technological Complexity:** Designing and building the DLT and the wiki interface with the required privacy, security, and permission features is complex. User-friendly interfaces are needed to make the technology accessible to a broad community.
- **Social Adoption and Cultural Shift:** Implementing such a system requires a cultural shift towards more explicit documentation, sharing, and collaborative validation of learning. Acknowledging resistance or apprehension from new users is essential.
- **Defining and Measuring Learning:** While LS offers discreet structures, capturing the nuances of embodied practice, experiential learning, and emergent outcomes remains challenging. Developing community-agreed metrics and evaluation processes is key. This revision adds a specific instance of the problem: **measuring syntropy** — a per-event, community-attested assessment of whether the event expanded the group's possibility space. Community evaluation is the instrument; the open work is converging on a shared scale or vocabulary for it.
- **The difficulty of articulation:** The subject matter is, by its nature, "super easy to use, hard as anything to explain." The protocol's interfaces and the Learning Journal must carry articulation that designers themselves struggle to produce; the system of justification field is a first answer, but the wiki must do real design work.
- **Trust propagation across communities:** The small-world model for how evidence and attribution travel across clustered local communities with few bridging nodes (§3.6.5) is not yet formalized. This is a research item: trust-decay functions across hops, hub recognition, and the governance of bridging roles.
- **Sustainability and Governance:** Establishing a sustainable model for the development and maintenance of the protocol and the Learning Journal, including decentralized governance mechanisms, is crucial.
- **Legal and Regulatory Landscape:** Navigating the evolving legal status of decentralized organizations and the use of DLT for non-financial purposes is necessary.

Future work will involve refining the protocol design, building and testing a minimum viable product, and engaging the LS community in a participatory design process. Research into integrating advanced privacy features, exploring different DLT platforms, developing user-friendly interfaces, formalizing the small-world trust-propagation model, and building community agreement on syntropy assessment will all be essential.

5. Conclusion

The Proof of Social Learning Protocol offers a compelling vision for leveraging DLT to support and amplify social learning within communities of practice like Liberating Structures. By using social learning events as the validation mechanism, the protocol aligns the technology with the community's core activities and values.

This revision adds the frame that makes the alignment visible: the events the protocol records are **organizational games**, the capacity they exercise is **design**, and the record itself is an **evidence archive of traces** — the community seeing and sensing its own system. Certification, when it comes, is not a product the protocol sells but a property the evidence acquires. And the protocol's economy is **syntropic** in a checkable sense: it rewards, records, and amplifies exactly those structures and practitioners that hold the water high — that leave every group they touch more organized, more capable, and with a larger possibility space than they found them.

The integrated Learning Journal can become a powerful tool for knowledge sharing, skill development, and fostering a regenerative economy of value based on collaboration, attribution, and the ethical distribution of

insights. While significant technical and social challenges remain, the potential for building a more transparent, equitable, and vibrant learning ecosystem makes this a worthwhile endeavor — and, by the protocol's own criterion, a syntropic one: the act of recording learning makes more learning possible.

Appendix A: Organizational Game Design — Key Concepts

Distilled from the "Press Start: Organizational Game Design" deck (Papa Pita, 2026). This is the practitioner's vocabulary underlying §3.

Organizational games. The broad category of socially constructed rule systems that organize a group's time, attention, information processing, and energy. Liberating Structures are the most mature, well-refined example; almost all human behavior can be read as game-like in this sense. The design question is how to distribute and clear away power so that games can be *good* — which is why the frame is games and game design rather than mere facilitation technique.

The metagame. The games that help you figure out *which* game to deploy. Two sides:

The Design Brief (situational assessment) — the five P's, battle-tested across ~200 help requests: - **Place** — the context, the arena the intervention finds itself in. - **Problem** — the tension driving change; the baseline; *why leaving things as they are is a horrible idea*. - **Purpose** — the ideal state, the "definition of awesome"; the speculative-fiction daydream that opens the possibility space. - **Participants** — those with influence and power, those with expertise, and those who will be affected; the players, whose play the game exists to create. - **Practicals** — constraints mapped by domain: *fixed* constraints (boundaries/walls), *governing* constraints (levers/knobs/gears), *enabling* constraints (resources; gravity is an enabling constraint). In practice: deadlines, cadences, availability windows.

The Design Storyboard (design and deploy) — the practitioner's most-used structure; "the alpha and omega." Fields per step: - **Agenda item** — what you'll call it *to a lay person* (they care about the artifact, not the mechanism). - **Goal** — what this step is for; the design brief jams into this slot. - **Invitation** — the question placed at the center of the conversation to steer it. - **Interaction pattern** — the rules: participation distribution, time, space, steps. - **Artifacts, materials, space** — maps (e.g., Ecocycle) and traces (e.g., Simple Ethnography, 25/10 Crowd Sourcing, What I Need From You). - **System of justification** — why these elements, why this design; how the choices line up into a coherent "slam dunk" on the goal.

Traces vs. agreements. Agreements are snapshots; traces are records of motion — artifacts that let you *see and sense the system*. The protocol's native data unit is the trace.

Syntropic flow. A flow of attention and information designed so that, as it moves, it leaves the group more organized and the possibility space larger. Emergence is not the absence of structure but structure designed to make emergence more probable — the opposite of the "tyranny of structurlessness."

The irony. "Super easy to use, hard as anything to explain" — which is why the field's answer is to get people talking to each other (the structures do the explaining), and why a shared evidence archive is not a luxury but a necessity for a practice that cannot narrate itself.

*Change log: v2.1 (2026-09-28) — Kitestring integration: §3.6.2 re-worked to reputation-first with a CENTRALIZED rubric / DECENTRALIZED seal split (criteria = one shared multi-dimensional Development Phases rubric: structures in use, people reached, diversity of settings, design-goal achievement; seal = user group's attestation; deck author holds no standing seat; demand-activated); community-governed pay-to-certify fee added — proceeds flow to the commons ("cash grab for the community"), never to the platform or deck author (§3.6.2, §3.6.3); collection/presentation independence added (§3.6.5); intended audience + value-first principle added (§3.6.4); §3.6.5 PENDING slot resolved by Mujo's Kitestring §07 finding (NAOMS trust edges unwired — credentials → trust-edges fix, 1–2 weeks). v2.0 (2026-08-23) — Organizational Game Design revision: added §1.1 (capacity as organizational game design), re-framed §3.1 as evidence archive with traces-vs-agreements, added first-class events to §3.2, extended §3.3 data model (roles, system of justification, traces, syntropy reflection), re-worked §3.5 progression, expanded §3.6 (evidence archive, certification as formalized capacity, market layer, first-class events, small-world trust propagation [pending model]), added §3.7 Syntropy as organizing principle, extended §4 (measuring syntropy, articulation difficulty, trust propagation), re-framed §5 conclusion, added Appendix A. races, syntropy reflection), re-worked §3.5 progression, expanded §3.6 (evidence archive, certification as formalized capacity, market layer, first-class events, small-world trust propagation [pending model]), added §3.7 Syntropy as organizing principle, extended §4 (measuring syntropy, articulation difficulty, trust propagation), re-framed §5 conclusion, added Appendix A.**